Problem Statement :
AI-Driven Fleet Management Optimization Platform

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We, as a technology-driven fleet management company, are seeking to develop an intelligent platform that leverages AI to optimize **vehicle utilization, reduce operational costs, and enhance driver safety.** Current solutions are fragmented, reactive, and lack predictive capabilities, making it difficult for fleet operators to proactively manage maintenance, route planning, and driver behavior.

The envisioned platform will provide real-time insights, predictive analytics, and automated recommendations to fleet managers, drivers, and maintenance teams. This will enable data-driven decisions, minimize vehicle downtime, and improve overall fleet efficiency.

Context

Fleet operators face challenges in tracking vehicle health, predicting maintenance needs, and optimizing routes, often resulting in unexpected breakdowns, increased costs, and safety risks.

Existing systems offer basic GPS tracking and manual reporting but lack advanced analytics, AI-driven predictions, and automated workflows.

There is a growing demand for sustainable, cost-effective, and safe fleet operations, especially as companies scale and regulatory requirements increase.

An AI-powered platform presents a strong opportunity to transform fleet management from reactive to proactive, driving business growth and customer satisfaction.

Requirements

- User Registration & Roles:
  - Users (fleet managers, drivers, maintenance staff) must register and be assigned roles with appropriate access.
- Vehicle & Driver Onboarding:
  - Ability to onboard vehicles (with telematics devices) and drivers, capturing essential details and compliance documents.
- Real-Time Vehicle Tracking:
  - Live GPS tracking of all vehicles, with status updates (idle, in transit, maintenance required, etc.).
- AI-Powered Predictive Maintenance:
  - The system analyzes telematics data to predict potential breakdowns and recommends maintenance schedules, reducing unplanned downtime.
- Route Optimization:
  - AI suggests optimal routes based on traffic, weather, and delivery schedules, aiming to reduce fuel consumption and improve delivery times.
- Driver Behavior Monitoring:
  - AI analyzes driving patterns (speeding, harsh braking, idling) and provides feedback to improve safety and efficiency.

Requirements

- Automated Alerts & Notifications:
  - Real-time alerts for critical events (maintenance due, route deviations, unsafe driving, etc.) sent to relevant users.
- Analytics Dashboard:
  - Comprehensive dashboard with KPIs (fuel usage, maintenance costs, driver scores, etc.) and AI-driven insights for continuous improvement.
- Integration Capabilities:
  - APIs to integrate with third-party logistics, ERP, and compliance systems.
- Admin Controls:
  - Admins can manage users, vehicles, AI rules, and data sources.
- Cost Management:
  - Track operational costs, with AI recommendations for cost-saving opportunities.
- Sustainability Tracking:
  - Monitor and report on fleet emissions, with AI suggestions for greener operations.

Artifacts for your consideration:

- Functional Requirements capturing (NFRs optional)
- List all the Actors
- Sequence diagram
- Use-case diagram
- Microservices – list all the services, explanation on how the services have been identified
- API Specification
- Class diagram
- Flow chart
- State diagram
- Activity diagram
- Object diagram
- ER Diagram
- Usage of guiding principles – SOLID, KISS, YAGNI

Thank you!